

(a) Show that if ϕ and $(\phi \rightarrow \psi)$ are tautologies, then ψ is a tautology.

Proof. Suppose that ϕ and $(\phi \rightarrow \psi)$ are tautologies.

Then for any truth assignment v we have $v(\phi) = T$ and $v(\phi \rightarrow \psi) = T$.

For $v(\phi \rightarrow \psi) = T$ to hold given that the antecedent, ϕ , is always true, it must be that the consequent, ψ , is always true.

Thus $v(\psi) = T$ □

(b) is it the case that if ϕ and ψ are tautologies, then $(\phi \rightarrow \psi)$ is a tautology?

Proof. Suppose that ϕ and ψ are tautologies.

Let v be any truth assignment. Then $v(\phi) = T$ and $v(\psi) = T$.

Thus $v(\phi \rightarrow \psi) = T$, as v respects the truth table for $\rightarrow$, which gives T on (T, T) .

Since v was arbitrary, $(\phi \rightarrow \psi)$ is a tautology, as desired. □

(c) It is the case that if $(\phi \rightarrow \psi)$ and ψ are tautologies, then ϕ is a tautology?

No. The claim is universally quantified over formulas, so we refute it by exhibiting particular formulas ϕ and ψ for which both hypotheses hold and the conclusion fails.

Counterexample. Let ϕ be the propositional variable p and let ψ be the formula $(p \rightarrow p)$.

ψ is a tautology. Let v be any truth assignment. The pair $(v(p), v(p))$ is either (T, T) or (F, F) , and the truth table for $\rightarrow$ gives T on both inputs, so $v((p \rightarrow p)) = T$. Since v was arbitrary, ψ is a tautology.

$(\phi \rightarrow \psi)$ is a tautology. Let v be any truth assignment. By the above, $v(\psi) = T$. The truth table for $\rightarrow$ gives F only on the input (T, F) ; since $v(\psi) = T$, this input does not occur, so $v((\phi \rightarrow \psi)) = T$. Since v was arbitrary, $(\phi \rightarrow \psi)$ is a tautology.

ϕ is not a tautology. By Theorem 2.2, any function from the propositional variables to $\{T, F\}$ extends to a truth assignment, so in particular there is a truth assignment v with $v(p) = F$. Then $v(\phi) = F$, so ϕ is not a tautology.

Thus $(\phi \rightarrow \psi)$ and ψ are tautologies while ϕ is not, so the answer is no. □

Remark. The second verification used nothing about ϕ : whenever $v(\psi) = T$, the fatal input (T, F) cannot occur. So if ψ is a tautology, then $(\phi \rightarrow \psi)$ is a tautology for *every* formula ϕ — a strengthening of part (b). The first hypothesis of (c) is thus a consequence of the second and places no constraint on ϕ whatsoever.

my version. Suppose $(\phi \rightarrow \psi)$ and ψ are tautologies. Let ϕ be a propositional variable. Since $(\phi \rightarrow \psi)$ is a tautology its range is just $\{T\}$. We see that the preimage of the truth assignment for $\rightarrow$ contains input $\{F, T\}$. So, it is not necessary for ϕ to be a tautology for $(\phi \rightarrow \psi)$ to be a tautology. □

(a) show that if ϕ and $(\phi \rightarrow \psi)$ are tautologies then ψ is a tautology.

Proof. Suppose ϕ and $(\phi \rightarrow \psi)$ are tautologies.

So let v be any truth assignment.

By definition $v(\phi) = T$ and $v((\phi \rightarrow \psi)) = T$.

We see that $v((\phi \rightarrow \psi)) = F$ if and only if $v(\phi) = T$ and $v(\psi) = F$.

But since $v((\phi \rightarrow \psi)) = T$ and $v(\phi) = T$, $v(\psi) \neq F$ so $v(\psi) = T$.

Since v was arbitrary, ψ is a tautology. $\square$

(b) is it the case that if ϕ and ψ are tautologies, then $(\phi \rightarrow \psi)$ is a tautology?

Proof. Suppose ϕ and ψ are tautologies.

So let v be any truth assignment.

By definition $v(\phi) = T$ and $v(\psi) = T$.

We see that $v((\phi \rightarrow \psi)) = F$ if and only if $v(\phi) = T$ and $v(\psi) = F$.

But since $v(\psi) = T$, we have $v(\psi) \neq F$, so $v((\phi \rightarrow \psi)) = T$.

Since v was arbitrary, $(\phi \rightarrow \psi)$ is a tautology. $\square$

(c) It is the case that if $(\phi \rightarrow \psi)$ and ψ are tautologies, then ϕ is a tautology?

Proof. this can only be proved through a counter example :/ $\square$